

Department of Chemical Engineering
National Institute of Technology Warangal
CH1309: Computational Methods in Chemical Engineering
Lab Handout 2

NOTE: Write all solutions in your Jupyter practice notebook, one Markdown heading per problem (e.g. "Problem 5"). Read every value with its units. Do not compare floating-point numbers with ==; use `np.isclose` or `np.allclose`. Give every plot axis a label with its unit. Run Kernel → Restart & Run; it must complete without errors. Data files (in `data/` next to your notebook): `calibration.csv`, `pipes.csv`, `reactor_runs.csv`. Antoine constants, $\log_{10} P_{\text{sat}}(\text{mmHg}) = A - B/(C + T(^{\circ}\text{C}))$: water (8.07131, 1730.63, 233.426); ethanol (8.20417, 1642.89, 230.30); benzene (6.90565, 1211.033, 220.79); toluene (6.95464, 1344.80, 219.48). $R = 8.314 \text{ J}/(\text{mol K})$; for water $\rho = 998 \text{ kg}/\text{m}^3$, $\mu = 0.001 \text{ Pa}\cdot\text{s}$.

Level 1.

- A laboratory study needs a set of temperatures at which to evaluate a property.
 - Create a NumPy array of 21 temperatures, equally spaced from 300 K to 500 K, with both endpoints included. State which array-creation function is appropriate when the number of points and both endpoints are known.
 - Compute the spacing between consecutive temperatures.
 - Print the number of points and the spacing.
- Twelve experiments are all carried out at atmospheric pressure, 101.325 kPa.
 - Create an array holding this pressure value twelve times, using `np.full`.
 - Compute and print the sum of the array.
- Consider the two lines `a = np.zeros(3, dtype=int)` followed by `a[0] = 7.9`.
 - Before running them, write down the value you expect `a[0]` to hold afterwards.
 - Run the two lines and print `a[0]`. Explain in one sentence why the result is what it is.
 - State the single change to the first line that stores 7.9 without loss.
- A reaction has pre-exponential factor $A = 2 \times 10^{11} \text{ s}^{-1}$ and activation energy $E_a = 65\,000 \text{ J/mol}$.
 - Using the Arrhenius equation $k = A e^{-E_a/RT}$, compute k at $T = 300, 350, 400, 450 \text{ K}$ in one vectorised expression.
 - Compute and print the ratio $k(400)/k(300)$.
- Six repeat measurements of a liquid density (kg/m^3) are 812.4, 813.1, 811.8, 812.9, 812.2, 813.4.
 - Compute the mean.
 - Compute the sample standard deviation (use `ddof=1`).
 - Find and print the index of the reading that lies furthest from the mean.
- The Reynolds numbers of seven pipes are
`Re = np.array([1500, 5200, 800, 96000, 3500, 42000, 2100])`.
 - Using a boolean mask, select the pipes that are turbulent ($Re > 4000$).
 - Count how many are turbulent, using `.sum()` on the mask.
 - Print the turbulent Reynolds numbers and the count.
- A table holds three experiments (rows) $\times$ three repeat temperature readings (columns):
(420.1, 421.5, 419.8), (455.3, 454.9, 455.8), (438.6, 439.2, 438.1).
 - Store the table as a 3×3 array.
 - Compute the mean of each experiment (one value per row).
 - Compute the mean of each repeat position (one value per column).
 - State which `axis=` value gives each of the two results.
- The file `calibration.csv` holds two columns: concentration (mmol/L) and absorbance.
 - Load the two columns with `np.loadtxt` (comma delimiter, skip the header row, unpack into two arrays).
 - Print the number of rows and the maximum absorbance.
 - Plot the measured data as points and the model line $A = 0.0521 C$ as a line. Label both axes with units and add a legend.
- The file `reactor_runs.csv` holds columns `run_id`, `T_K`, `catalyst`, `feed_L_per_min` and `conversion`.
 - Load the file with `pandas.read_csv`.
 - Compute the mean conversion over all runs.
 - Count how many runs used catalyst "B".
 - Print both results.

Level 2.

- On a grid of temperatures from 60 to 110 $^{\circ}\text{C}$ in steps of 0.5 $^{\circ}\text{C}$, find where water's vapour pressure first exceeds 500 mmHg.
 - Build the grid and compute water's vapour pressure on it (Antoine constants in the table above).
 - Using a boolean mask, find the first grid temperature at which $P^{\text{sat}} > 500 \text{ mmHg}$. Do not use a loop.
 - Print that temperature.
- Six pipes have the following Reynolds numbers and diameters (m):

$$Re = (2400, 3100, 5600, 2900, 1500, 3800)$$
$$D = (0.025, 0.075, 0.040, 0.030, 0.020, 0.060)$$

- Using one compound boolean mask, select the pipes that are transitional ($2100 \leq Re \leq 4000$) and have diameter below 0.05 m. Use `&` with parentheses.
- Print the indices of the selected pipes.
- For each pipe you rejected, note in a comment which condition it failed.

12. For four heat-exchanger cases the terminal temperature differences (K) are:

$$\Delta T_1 = (40, 25, 30, 20.000001) \quad \Delta T_2 = (20, 25, 10, 20)$$

The log-mean temperature difference is $(\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2)$, which is 0/0 when the two differences are equal.

- Use `np.isclose` to find the cases where ΔT_1 and ΔT_2 are equal.
 - Compute the LMTD for all four cases with `np.where`, substituting the arithmetic mean where the two differences are equal.
 - Print the four LMTD values and the number of cases that used the fallback. Note what happens for the pair 20.000001 vs 20.
13. A rate constant is $k = 0.045 \text{ s}^{-1}$ at 320 K and $k = 1.32 \text{ s}^{-1}$ at 380 K.
- Write the two-point Arrhenius expression $E_a = R \ln(k_2/k_1)/(1/T_1 - 1/T_2)$ in code.
 - Compute E_a and print it in kJ/mol.
14. Let `M = np.array([[2.,1.],[0.5,3.]])` and `v = np.array([4.,2.])`.
- By hand, write down what `M * v` and `M @ v` each produce.
 - Compute both in code and confirm your predictions.
 - In one sentence, state the difference between the two operators.
15. The file `pipes.csv` holds pipe_id and the columns D , v , ρ , μ for ten pipes.
- Load the file and compute the Reynolds number $Re = \rho v D / \mu$ for all ten pipes.
 - Report the pipe id and value of the largest and smallest Reynolds number.
 - Count how many pipes are laminar ($Re < 2100$), transitional ($2100 \leq Re \leq 4000$) and turbulent ($Re > 4000$).

Level 3.

16. For every pipe in `pipes.csv` (roughness $\varepsilon = 4.5 \times 10^{-5} \text{ m}$, length 10 m), the Darcy friction factor is $f = 64/Re$ when $Re < 2100$, and otherwise follows the Haaland equation $1/\sqrt{f} = -1.8 \log_{10}[(\varepsilon/D/3.7)^{1.11} + 6.9/Re]$.
- Compute f for all pipes with a single `np.where` (no loop), then the pressure drop for each pipe:

$$\Delta P = f (L/D) \rho v^2 / 2.$$

- Report the pipe with the largest pressure drop and its ΔP .
 - Verify one laminar and one turbulent friction factor by hand.
17. Build a 3×501 array of Antoine vapour pressures for water, ethanol and benzene over 60 to 110 °C (0.1 °C steps), using broadcasting.
- Put the three sets of constants as (3,1) columns and the temperature grid as a (1,501) row.
 - For each component, find the first grid temperature at which P^{sat} exceeds 760 mmHg.
 - Report the maximum absolute error against the known normal boiling points 100, 78.37 and 80.10 °C.
18. An orifice meter gives $Q = C_d A \sqrt{2\Delta P/\rho}$ with $A = 5 \times 10^{-4} \text{ m}^2$ and $\rho = 998 \text{ kg/m}^3$. The discharge coefficient and pressure drop are uncertain: $C_d \sim N(0.61, 0.01)$ and $\Delta P \sim N(25\,000, 1000) \text{ Pa}$.
- Seed a generator with 42; draw 100 000 samples of C_d first, then of ΔP ; compute Q for all samples in one vectorised expression.
 - Report the mean, standard deviation and relative standard deviation (%) of Q .
 - Compare the relative standard deviation with the linear-propagation estimate

$$\sqrt{(\sigma_{C_d}/C_d)^2 + (\frac{1}{2} \sigma_{\Delta P}/\Delta P)^2},$$

and explain the factor $\frac{1}{2}$.

19. Save the following eight-row file as `dirty.csv` and load it: `t_min, T_K / 0, 350.2 / 5, / 10, 352.8 / 15, 353.9 / 20, / 25, 356.1 / 30, 357.0 / 35`, (three temperature values are missing).
- Load the file with `np.genfromtxt` so the gaps become `nan`.
 - Report how many temperature values are missing, the nan-aware mean of the temperature, and the number of complete rows.
 - Explain in one sentence why a plain `.mean()` fails on this column.
20. Generate a synthetic calibration data set and test it against a specification.
- With a generator seeded 0, take $C = 0$ to 10 mmol/L (11 points) and set $A = 0.0521 C + \text{noise}$, $\text{noise} \sim N(0, 0.004)$.
 - Compute the largest absolute deviation of the noisy data from the noise-free line, and state whether it satisfies a 0.01 AU specification.
 - Explain in one sentence why using the same seed guarantees the same verdict on any computer.